

Nathan David Obeng-Amoako, E.I.T.

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RESEARCH INTERESTS

- Transit Signal Priority
- Transportation Safety Analysis
- Intelligent Transportation Systems
- Multimodal Corridor Planning
- Bicycle Network Analysis
- Geospatial Methods in Transportation

EDUCATION

Doctor of Philosophy in Civil Engineering (Transportation) Sep 2023 – Present

Northeastern University, Boston, MA (GPA: 4.00/4.00)

Research Advisor: Professor Peter Furth

Master of Science in Civil Engineering (Transportation) Sep 2023 – Apr 2025

Northeastern University, Boston, MA (GPA: 4.00/4.00)

Bachelor of Science in Civil Engineering Aug 2018 – Sep 2022

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

Recognition: Best Graduating Student, Class of 2022

PROFESSIONAL EXPERIENCE

Transportation Engineering Intern, Fehr & Peers, Oakland, CA Apr 2026 – Jun 2026

- Supported transit signal priority and complete streets projects for state/municipal clients
- Contributed to metro station area studies and transit-oriented development analysis in the East Bay
- Developed federal grant materials for the City of Palo Alto under the OBAG 4 funding cycle
- Conducted GIS and active transportation analysis for corridor and safety planning projects

Program Assistant, Sustainable Transportation Summer Academy, Delft, Holland Jun 2024 – Aug 2026

- Provided logistical, academic, and evaluative support to students over three summers
- Assisted in teaching Vision Zero safety principles to 25 Northeastern University students
- Explored car-free and car-lite zone strategies to enhance active/public transportation

Assistant Transportation Engineer, BHM Construction International UK Ltd. Oct 2022 – Jul 2023

- Conducted over 20 quality control tests daily on construction materials to ensure strict standard compliance
- Supported site reporting and coordination with a team of 5 engineers to improve workflow efficiency

Civil Engineering Trainee, China Henan International Corporation Ltd. Sep 2021 – Dec 2021

- Supervised 15+ artisans daily to ensure strict adherence to project specifications and timelines
- Contributed to precise structural detailing for reinforced concrete components

SPONSORED RESEARCH PROJECTS

Project: Bicycle Level of Traffic Stress at Intersections (xLTS) **Jun 2024 – Aug 2025**

Funding Agency: United States Department of Transportation (USDOT)

- Assessing the impact of high-stress crossings on bike network connectivity
- Developing algorithms and data structures for traffic stress network analysis

Project: Accessible Bus Stop Design in the Presence of Bike Lanes

Sep 2023 – Jun 2024

Funding Agency: Massachusetts Department of Transportation (MassDOT)

- Studied bus rider and bicyclist behavior at floating and constrained bus stops
- Proposed design improvements to reduce bus-rider-and-bicyclist conflicts

PEER-REVIEWED JOURNAL PUBLICATIONS

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|------|--|
| 2025 | 1. Peter Furth, Ray Saeidi-Razavi, Nathan David Obeng-Amoako and Milad Tahmasebi. TSP-Friendly Underlying Traffic Signal Control: An Essential Complement of Transit Signal Priority. <i>Future Transportation</i> (2025), doi: https://doi.org/10.3390/futuretransp5040160 |
| | 2. Joshua Kofi Asamoah, Blessing Agyei Kyem, Nathan David Obeng-Amoako , Armstrong Aboah, SAAM-ReflectNet: Sign-Aware Attention-Based Multitasking Framework for Integrated Traffic Sign Detection and Retroreflectivity Estimation, <i>Expert Systems With Applications</i> (2025), doi: https://doi.org/10.1016/j.eswa.2025.128003 |

Submitted and under review:

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| 2026 | 3. Peter G. Furth, Nathan David Obeng-Amoako , Armin Akhavan, Bitra Sadeghinassr, Ray Saeidi, Luis Miranda-Moreno, Christine Théberge-Barrette, Paul Moser, and Nick Ferenchak. Intersections as Barriers to Low-Stress Cycling: Intersection Level of Traffic Stress and Dual-Graph Network Modeling. <i>Journal of Transport Geography</i> . |
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RESEARCH REPORTS

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| 2024 | 1. Eleni Christofa, Chengbo Ai, Peter Furth, Yu-Min Yang, Dewan Tanvir Ahammed, and Nathan David Obeng-Amoako . (2024). <i>Accessible Bus Stop Design in the Presence of Bike Lanes</i> . Massachusetts Department of Transportation (MassDOT), Office of Transportation Planning. Report No. 24-060. Retrieved from https://rosap.ntl.bts.gov/view/dot/79327 |
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JOURNAL REVIEWS

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| 1. Journal of Transportation Engineering, Part A: Systems, ASCE | 2026 – Present |
| 2. Transportation Research Interdisciplinary Perspectives, Elsevier | 2025 – Present |
| 3. Expert Systems with Applications, Elsevier | 2025 – Present |
| 4. Transportation Research Record, SAGE Publications | 2024 – Present |

APPOINTMENTS AND ADVISORY ROLES

Member, Massachusetts Bicycle and Pedestrian Advisory Board, MassDOT

Jan 2026 – Present

- Appointed by Governor Maura T. Healey to advise on state-wide bicycle and pedestrian infrastructure policies and programs
- Provide guidance on transportation infrastructure, safety, and accessibility initiatives

RELEVANT SOFTWARE AND SKILLS

- **Python, R, MATLAB** machine learning, deep learning, advanced data analysis
- **PTV Vissim/Vistro** traffic flow simulations, traffic signal analysis, traffic impact analysis
- **Synchro** traffic flow modeling, traffic signal timing optimization
- **ArcGIS** spatial data mapping and analysis for informed decision-making
- **Gephi** network data analysis and visualization
- **AutoCAD** precision drafting and technical drawings, 2D and 3D
- **HTML/CSS/JavaScript** developing web applications
- **Minitab, MS Excel** data visualization, statistical computing, data organization

NOTABLE DISTINCTIONS AND ACHIEVEMENTS

- Grant Recipient, New England African Business Expo, Worcester, MA **May 2025**
- Civil & Environmental Engineering Distinguished Fellowship Award, Northeastern Univ. **Aug 2023**
- Associated Consultants Ltd. Award for Best Graduating Student, KNUST **Jul 2023**
- Provost Award for Best & Excellent Students in College of Engineering, KNUST (x4) **2019 – 2022**
- Academic Excellence Award, Ghana Engineering Students' Association, Ghana **Sep 2022**
- Global Finalist, Unilever Future Leaders League Business Pitch Competition, U.K. **Jul 2022**
- Scholar, Tullow Oil Scholarship Scheme, Ghana **Apr 2019**
- Semifinalist, National Science and Maths Quiz (NSMQ), Ghana **Jul 2018**

SELECTED CONFERENCE PRESENTATIONS

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| 2026 | <ol style="list-style-type: none">1. Furth, P., & Obeng-Amoako N. D. (2026, May). <i>Why Johnny and Mary Can't Ride to School: Identifying Barriers to Safe Bicycle Access</i>. Boston Area Research Initiative (BARI) Conference, Boston, MA.2. Yang, Y., Ahammed, D. T., Obeng-Amoako, N. D., Christofa, E., Ai, C., & Furth, P. (2026, January). <i>Community-Engaged Informed Floating Bus Stop Guidelines</i> [Poster]. Transportation Research Board Annual Meeting, Washington, DC. |
| 2025 | <ol style="list-style-type: none">3. Obeng-Amoako, N. D., & Furth, P. (2025, May). <i>Underlying Traffic Signal Control: The Key to Effective Transit Signal Priority – A Case Study of MBTA Bus Route 39 along South Huntington Avenue, Boston, MA</i> [Poster]. MassDOT Transportation Innovation Conference, Worcester, MA.4. Yang, Y., Ahammed, D. T., Obeng-Amoako, N. D., Christofa, E., Ai, C., & Furth, P. (2025, June). <i>Design Guidelines for Accessible Floating Bus Stops</i>. International Conference on Transportation and Development (ASCE), Glendale, AZ. |
| 2024 | <ol style="list-style-type: none">5. Yang, Y., Ahammed, D. T., Obeng-Amoako, N. D., Christofa, E., Ai, C., & Furth, P. (2024, April). <i>Accessible Bus Stop Design in the Presence of Bike Lanes</i> [Poster]. MassDOT Transportation Innovation Conference, Worcester, MA. |

- 2023 6. **Obeng-Amoako, N. D.**, Senaya, A. L., Kassim, B., Owusu-Ansah, F., Damoah, E., Worclachie, S., & Tutu, K. A. (2023, April). *Modification of Bitumen with Waste Materials for Enhanced Aggregate Retention in Surface-Dressed Roads*. 3rd IRF Africa Regional Congress & Exhibition, Accra, Ghana.

LEADERSHIP EXPERIENCE

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| Secretary, Ghana Transportation Professionals Forum (GTPF) | Feb 2026 – Present |
| <ul style="list-style-type: none"> • Maintain membership records and coordinate executive board communications • Support event logistics and engagements with Ghana transportation stakeholders | |
| Committee Member, Career Mixer Committee, CEE Dep't, Northeastern Univ. | Sep 2025 – Feb 2026 |
| <ul style="list-style-type: none"> • Organizing networking events connecting students with industry • Coordinating logistics for career dev't workshops and employer engagements | |
| Executive C'ttee Member, CEE Graduate Student Council, Northeastern Univ. | Feb 2024 – Present |
| <ul style="list-style-type: none"> • Brainstorming and implementing strategies to enhance graduate student experience • Serving as a liaison between the departmental chair and the graduate student body | |
| Department Rep., College of Engineering. PhD Council, Northeastern Univ. | Jun 2024 – Aug 2025 |
| <ul style="list-style-type: none"> • Meeting with advisor to discuss novel ways of improving student experience • Organizing weekly coffee hours and semesterly happy hours for PhD students | |
| Communications Director, Young Professionals and Youth Coalition, Ghana | Jan 2021 – Dec 2021 |
| <ul style="list-style-type: none"> • Social media management to curate consistent communications content • Organization of in-person and virtual engagements with members and partners | |
| President, Civil Engineering Students' Association, KNUST | Feb 2020 – Sep 2021 |
| <ul style="list-style-type: none"> • Coordination of fundraising events to secure administrative equipment • Establishment of academic support scheme for underachieving students | |

TEACHING EXPERIENCE

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| Northeastern University, Boston, MA | |
| Spring 2026 | Graduate Teaching Assistant, Energy Systems: Science, Technology, and Sustainability <ul style="list-style-type: none"> • Provided grading support for coursework on energy technologies and sustainability • Gave an invited lecture on Transportation Energy Systems in a Developing World |
| Spring 2026 | Graduate Teaching Assistant, Transportation Systems Analysis and Planning <ul style="list-style-type: none"> • Supported students during office hours with questions on design software (Synchro) • Evaluated assignments on intersection analysis, signal timing, and street design principles |
| Fall 2025 | Graduate Teaching Assistant, Highway Design <ul style="list-style-type: none"> • Provided grading support for homework assignments • Assisted students during office hours, addressing questions about design principles |

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| Fall 2025 | Graduate Teaching Assistant, Transportation Systems Analysis and Planning <ul style="list-style-type: none"> • Evaluated coursework and provided constructive feedback on analytical assignments • Supported students' understanding of transportation system planning frameworks |
| Fall 2023 | Course Instructor, Recitation for Statics and Solid Mechanics <ul style="list-style-type: none"> • Tutored fundamental concepts of Newtonian physics and solid mechanics • Evaluated students' test performance and offered comprehensive feedback |

PROFESSIONAL AFFILIATIONS

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| • American Society of Civil Engineers (ASCE) | Feb 2025 – Present |
| • Boston Society of Civil Engineers Section (BSCES) of ASCE | Feb 2025 – Present |
| • New England Intelligent Transportation Society (NEITS) | Sep 2023 – Present |
| • Institute of Transportation Engineers (ITE) | Jan 2024 – Present |

VOLUNTEER EXPERIENCE & COMMUNITY INVOLVEMENT

- **Judge**, Massachusetts Science and Engineering Fair, Worcester (April 2025)
- **Cycling Activist**, Boston Cycling Union, Boston, MA (Feb 2025 – Present)
- **Executive Vice President**, Student Energy Society, KNUST (Jan 2022 – Oct 2022)
- **Community Outreach Coordinator**, Nat'l Society of Black Engineers, KNUST (Jan 2022 – Sep 2022)
- **Project Coordinator**, Engineers Without Borders, KNUST (Aug 2021 - Jul 2022)

BLOG POSTS

1. "Ghana is Moving, but Barely: The Case for a Smarter Public Transportation Strategy." MyJoyOnline, April 2026. <https://www.myjoyonline.com/ghana-is-moving-but-barely-the-case-for-a-smarter-public-transportation-strategy/>
2. "Replacing the Reid Overpass with an At-Grade Intersection (Cambridge, near BU Bridge)." Co-authored with Peter G. Furth. January 2026. <https://peterfurth.sites.northeastern.edu/2026/01/06/replacing-the-reid-overpass-with-an-at-grade-intersection-cambridge-near-bu-bridge/>
3. "PhD Student Reflects on the Sustainable Transportation Dialogue of Civilizations in Holland." Northeastern University College of Engineering News, August 2024. <https://coe.northeastern.edu/news/a-reflection-on-the-sustainable-transportation-dialogue-of-civilizations-course-in-holland/>
4. "Bus-Friendly Traffic Signals Can Reduce Bus Delay By 90% on South Huntington Avenue." Co-authored with Peter G. Furth. April 2024. <https://peterfurth.sites.northeastern.edu/2024/06/04/bus-friendly-traffic-signals-can-reduce-bus-delay-by-90-on-south-huntington-avenue/>

REFERENCES

Available upon request